Synopsis - Mandatory 2

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Research Question

Does removing non-anatomical artifacts from chest X-rays improve pneumonia classification performance?

Sub-questions:

- How can we detect non-anatomical artifacts?
- What is the most effective method of removing the non-anatomical artifacts?

Method

The project follows Nunamaker et al.'s multi-methodological systems development approach, combining data exploration, iterative model building, and experimental evaluation. The dataset consists of chest X-ray images labeled as NORMAL or PNEUMONIA. Because the original split was suboptimal with a very small validation set, the data was reorganized into an 80/10/10 distribution to ensure consistent and comparable evaluation across experiments.

Initial analysis included class distribution visualizations and inspection of sample images, which revealed the presence of non-anatomical artifacts such as laterality markers (e.g., "R/L") and imaging system annotations. A preprocessing pipeline was developed: images are converted to grayscale with three channels, padded to preserve aspect ratio, resized to 224×224 pixels, and normalized. During training, light augmentation (small rotations, translations, and zoom) is applied to improve generalization, and a WeightedRandomSampler is used to balance NORMAL and PNEUMONIA representation in each batch.

Experimental Design

A baseline CNN is trained on the original dataset containing non-anatomical artifacts. An identical model is then trained on a cleaned dataset where artifacts are removed. Architecture, hyperparameters, and training procedures are kept constant across both conditions to ensure a controlled comparison. Cleaning approaches under consideration range from simple corner masking and threshold-based cropping to OpenCV-based artifact detection with inpainting and pretrained U-Net lung segmentation model.

Choice of Model

Transfer learning with ResNet-18 and DenseNet-121 was explored but discarded to keep training feasible on available hardware. Instead, custom CNNs with additional hidden layers and more nodes per layer are trained from scratch. Hyperparameter tuning (learning rate, dropout, batch size, weight decay) is performed via random search, selecting the configuration with the highest AUC.

Performance Metrics

Because the dataset is imbalanced (≈73% pneumonia, 27% normal), accuracy is reported only as a secondary metric — a naive classifier predicting pneumonia for every image would already reach 73% accuracy. The primary metric is macro F1-score, which balances precision and recall while weighting both classes equally. Pneumonia recall (sensitivity) is emphasized because false negatives risk missed diagnoses, while pneumonia precision limits false positives and normal-class recall (specificity) ensures healthy patients are not over-classified as sick. PR-AUC is included as it is well-suited for imbalanced tasks, and a confusion matrix is used to interpret prediction behavior. ROC curves and threshold tuning support the final evaluation.

Expected Contribution

The study aims to determine whether removing non-anatomical artifacts produces measurable gains in classification performance, and whether deep learning models rely on such artifacts as predictive shortcuts rather than learning genuine anatomical features.

Development process

1. Load data
2. Split data (80/10/10)
3. Data viz and exploration
4. Class imbalance found
5. Preprocessing (Pad, resize, grayscale, zoom rotate)
6. WeightedRandomSampler(Handles imbalance)
7. Make Custom CNN
8. Train baseline for 15 epochs
    a. Evaluate loss,acc,prec,rec on train and val set
    b. Plot the above
9. Hyperparameter Tuning via Random search
    a. Learning rate, dropout, batch_size, weight_decay searched
    b. Keep the with the highest AUC(Area under curve)

10. Train best model
11. Evaluate on test set, compare models
    a. Confusion matrix
    b. ROC Curves
    c. Threshold tuning